

Why The Second Research Direction is Impractical For Now

The research direction (number 2) focusing on quantum-accelerated VRP preprocessing, specifically the All-Pairs Shortest Path (APSP) and Single-Source Shortest Path (SSSP) problems, is deemed impractical. The following points outline the key issues.

Theoretical Quantum Algorithms

The proposed quantum distributed algorithms for APSP, based on research by Izumi and Le Gall [1], are considered purely theoretical and lack a practical implementation in current quantum settings [1]. Similarly, the idea of using quantum dynamic programming for the Bellman-Ford algorithm is based on a paper by Caroppo et al. [2], which frames the approach as an open problem rather than a practical solution framework [2]. This makes these methods irrelevant for a research project that requires a tangible, implementable solution.

Algorithmic Obsolescence

Current classical algorithms are advancing rapidly. The idea of using Grover’s algorithm within Dijkstra’s to find minimum values is based on a greedy, probabilistic approach. This method is already at risk of becoming outdated. A new deterministic SSSP algorithm [3], which is faster than Dijkstra and uses a priority queue instead of a matrix data structure, has recently been published. This highlights the need to constantly keep up with fast-paced classical research, making a quantum approach potentially obsolete before it’s developed.

Effective Classical Alternatives

Highly effective classical solutions already exist. For example, Google’s OR-Tools provides a fast VRP solution using heuristic methods, which are already considered excellent in this domain. The availability of such robust classical tools reduces the need for a novel, and currently impractical, quantum approach.

References

- [1] Taisuke Izumi and François Le Gall. Quantum distributed algorithm for the all-pairs shortest path problem in the congest-clique model. In *Proceedings of the 38th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 84–93, 2019.
- [2] Susanna Caroppo, Giordano Da Lozzo, Giuseppe Di Battista, Michael T. Goodrich, and Martin Nöllenburg. Quantum speedups for polynomial-time dynamic programming algorithms. arXiv preprint arXiv:2507.00823, 2025.
- [3] R. Duan, J. Mao, X. Mao, X. Shu, and L. Yin, “Breaking the Sorting Barrier for Directed Single-Source Shortest Paths,” arXiv:2504.17033 [cs.DS], 2025.